

# Yuchen Wang

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## Education

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| <b>University of Illinois Urbana-Champaign</b> , Computer Science                                                               | Champaign, IL, USA<br>2025 – 2027 |
| <b>Peking University</b> , Intelligent Science and Technology (Artificial Intelligence)<br>• Zhi Class (Elite Program), Top 10% | Beijing, China<br>2021 – 2025     |

## Experience

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| <b>Alibaba U.S.</b> , Research Intern — Agentic LLM Post-Training                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Sunnyvale, CA, USA<br>May 2026 – Dec 2026<br>8 months |
| <ul style="list-style-type: none"> <li>Contributed to Occamy-1.0 with Alibaba's Accio Team, focusing on execution-grounded data and post-training infrastructure. The model builds on Qwen3.6-35B-A3B with Marathon (SFT + HDPO) and Sprint (SFT) experts, model soup, and post-merge SAO.</li> <li>Built a verifier-gated, multi-harness data and training pipeline with immutable provenance, token-exact proxy/TiTO replay, state reconstruction, context-segment lineage, and loss masks. Under the matched Claw-Eval T/C protocol, the model uses 19.5% fewer tokens per trajectory and 46.4% less trace wall time than the starting checkpoint; timeout rate falls from 9.88% to 2.18%.</li> <li>The latest technical report records 82.2 Claw-Eval average (71.4 Pass<sup>3</sup>), 49.16 WildClaw-Bench, 27.6% AutomationBench strict pass rate, 48.1 OfficeQA Pro, and 59.0 Terminal-Bench 2.1. The multi-harness training stack is publicly released as Dressage.</li> </ul> |                                                       |
| <b>Freedo Technology</b> , Research Intern — 3D Vision and Generative AI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Beijing, China<br>May 2025 – Aug 2025<br>4 months     |
| <ul style="list-style-type: none"> <li>Built an end-to-end depth-conditioned ControlNet pipeline for 3D reconstruction from noisy point clouds, improving geometric fidelity from 32.7% to 85.3%.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                       |

## Open Source Projects

### Open-Source LLM Systems Engineering

Upstream reliability and performance work across FlashInfer, vLLM, NVIDIA NeMo, Megatron-LM, Hermes Agent, SGLang, verl, vLLM/Vime, and ModelScope.

- Numerics and training: FlashInfer #4984 fixes FP8 KV prefill calibration (40 B200 regression tests passed). Contributions also cover Muon and hybrid Mamba/MoE correctness, sequence packing, NCCL startup, non-finite RL logprobs, GDN/Muon clipping and routing, skipped-tensor weight checks, and FSDP validation.
- Performance and scale: vLLM #54699 halved TP2 peak allocation from 7.88 to 3.94 GiB and enabled real 120B load, generation, and 3/3 sleep-wake cycles in earlier validation; current-head targeted GPU tests passed 112/112 on 8×B200. NeMo RL #3943 delivered 4.44–5.31× controlled Ray transfer speedup with 0.1–0.3 MB driver RSS deltas; Megatron-LM work fused GatedDeltaNet normalization and added Mamba recompute.
- Agent runtime: Hermes #100693 and #102549 repair schema-reference argument coercion and ownership-checked SSH backend cleanup (22 and 14 targeted tests passed). Co-authored NeMo Gym #2726 for pickle-safe cross-process failures; SGLang #38063 adds cold-JIT startup diagnostics.

## Projects

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### **CineFlow — Dependency-Driven Parallel Video Generation**

The first dependency-driven video-diffusion inference system, built around semantic DAGs, runtime trajectory fusion, online edge pruning, and critical-path scheduling without retraining.

- Analyzed 1K+ MovieGenBench prompts; more than 68% exhibit non-linear dependencies.
- Across Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on 8×H100, achieved 1.7–5.5× end-to-end speedup, 1.30–2.02× lower P90 latency, and 5.4–17.3% higher VBench overall; pruning added nearly 7% throughput and improved visual coherence by 6.7–14.6%.

### **Dynamic Prefill Optimization via Adaptive Online Packing**

AIMD control with live p95 TTFT feedback and greedy/DP prompt packing.

- Achieved up to 20% lower TTFT on production-style DynamoLLM traces.

### **FlashAttention-style CUDA Optimization for GPT-2**

Tiled online softmax, memory-efficient attention, and kernel fusion.

- Reduced HBM traffic by roughly 10× and achieved up to 9% end-to-end speedup.

### **Reinforcement Learning for Legal Reasoning**

Advised by Prof. Yansong Feng at Peking University; Zero-RL → distilled-CoT SFT → GRPO for structured legal multiple-choice reasoning.

- Reached 57.6% accuracy, outperforming SFT-only and RL-only baselines.

## Skills

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- Python, C++, CUDA, PyTorch, Ray, FSDP, 3D vision, point clouds, ControlNet, diffusion transformers, Megatron-LM, NeMo RL, vLLM, SGLang, distributed training, and profiling